C2.2 Neural signalling (HL)

NAME	DATE	TOTAL
		_____ / 40

About 40 minutes, or one section at a time. The number of marks shows how many separate points to make.

Section A: Recall

8 MARKS

1 STATE the approximate threshold potential of a neuron. [C2.2.8] [1]

2 STATE the ion that enters the axon during depolarization. [C2.2.8] [1]

3 NAME one inhibitory neurotransmitter. [C2.2.13] [1]

4 STATE where ion channels are concentrated in a myelinated axon. [C2.2.11] [1]

5 OUTLINE why consciousness is described as an emergent property. [C2.2.16] [2]

6 STATE two stimuli that open the ion channels of pain receptors. [C2.2.15] [2]

Section B: Understanding

16 MARKS

7 EXPLAIN how voltage-gated channels cause depolarization and repolarization. [C2.2.8] [4]

8 EXPLAIN how local currents propagate an action potential along an axon. [C2.2.9] [3]

9 **EXPLAIN** how neonicotinoids affect synapses in insects. [C2.2.12] [3]

10 **EXPLAIN** how an inhibitory neurotransmitter produces an inhibitory postsynaptic potential. [C2.2.13] [3]

11 **EXPLAIN** how summation decides whether a postsynaptic neuron fires. [C2.2.14] [3]

Section C: Working with evidence

10 MARKS

All data in this section are invented for practice. The patterns are realistic.

12 [C2.2.11, C2.2.9]

The speed of nerve impulses was measured in axons of different diameters at 37 °C.

AXON DIAMETER / MM	1	2	5	10	15	20
MYELINATED: SPEED / M S^{-1}	6	12	30	60	90	120
UNMYELINATED: SPEED / M S^{-1}	1.0	1.4	2.2	3.2	3.9	4.5

a DESCRIBE the relationship between diameter and speed in myelinated axons. [2]

b CALCULATE how many times faster a myelinated axon is than an unmyelinated one, at 10 µm. [1]

c EXPLAIN the difference in speed between myelinated and unmyelinated axons. [2]

d CALCULATE the time an impulse takes to travel 1.5 m along a myelinated axon 15 µm wide. Give your answer in ms. [2]

e SUGGEST why squid, which have no myelin, have axons up to 1 mm wide that control escape. [2]

f STATE why the temperature was kept the same. [1]

Section D: Longer answer

6 MARKS

13 EXPLAIN how touching a hot surface leads to the perception of pain. [C2.2.15, C2.2.8] [6]